Adepu Sanjana

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2026	8.28
Narayana Junior College	Class XII (SBIE)	2022	98.60%
ZPBHS	Class X (SSC)	2020	100.00%

WORK EXPERIENCE

Full Stack Developer Intern | Rengy

- Owned the development of a backend microservice within a **Fastify**, **MongoDB** microservice architecture.
- Developed production-ready APIs, **WebSocket** integration, authentication, reusable OTP services, and dynamic payment PDF generation.
- Built secure **AWS S3** file upload pipelines with use-case-specific validation for images, videos, and documents.
- Refactored backend services by eliminating duplicate logic, introducing reusable abstractions, and optimizing MongoDB queries without changing API contracts.
- Collaborated with frontend, mobile, and QA teams to resolve cross-platform issues and deliver production-ready features using **React**, **TanStack Query**, and **shadcn/ui**.

PROJECTS

E-Commerce:

- Built a full-stack MERN e-commerce web application with JWT-based authentication, product and category management, shopping cart, checkout, and order management.
- Implemented admin dashboard functionality for managing products, categories, and orders, along with protected routes and role-based access.
- Developed product search, category filtering and responsive UI for an improved shopping experience.
- **Tech Used:** React.js, React Router, Node.js, Express.js, MongoDB, Mongoose, JWT, Axios, Bootstrap.

ERF AI:

- Built an AI-assisted company registration system using Gemma 2B via Ollama for structured ERF data extraction.
- Implemented RAG-based missing-field recovery from JDs using 2048-dimensional embeddings; evaluated retrieval across 10 ERF-JD test cases.
- Tracked LLM token usage, tuned decoding parameters for deterministic extraction, and debugged APIs using FastAPI Swagger.
- **Tech Used:** React.js, FastAPI, SQLite, Ollama, Gemma 2B, RAG, Embeddings

IITH-NCC Website

- Collaborated with a senior to develop the official NCC website of IIT Hyderabad, contributing to frontend development and content management.
- Curated and uploaded training schedules, achievements, gallery updates, and recruitment guidance to provide informative content for NCC cadets.
- **Tech Used:** React.js, Tailwind CSS, AOS Animations.

POSITIONS OF RESPONSIBILITY

Senior Under Officer - National Cadet Corps, IITH

- Equally responsible as a club head for leading and training cadets in both drill and theoretical aspects of NCC.
- Conduct weekly sessions for nearly 50 cadets, collaboratively planning and executing training alongside other rank holders
- Providing mentorship, guidance, and administrative support to fellow cadets, ensuring smooth execution of NCC-related activities and events.

Technical Coordinator - Office of Career Services (OCS)

- Maintain and upgrade the OCS website along with the team, handling large student databases and codebases to resolve student issues efficiently.

Senior Core Member - Lambda (Software Development Club, IITH)

- Managed coordination between club heads and core members.
- Assigning tasks, clarifying requirements, and ensuring timely delivery.

Events Core Member - Prakriti Club

- Designed digital posters and supported event coordination for campus campaigns.

SKILLS

- **Frontend:** ReactJS (Vite), Tailwind CSS, Material UI (MUI), TanStack Query, Axios
- **Backend:** Node.js, Express.js, REST APIs, WebSockets, Microservices, MongoDB, MySQL, JWT, bcrypt
- **Programming Languages:** C++ (STL, DSA), JavaScript
- **Core Subjects:** DBMS, Computer Networks, OOPs, Operating Systems
- **Soft Skills:** Leadership, Time Management, Curiosity to Learn through Application, Multitasking, Problem-Solving
- **LLM Engineering:** LLM Fundamentals, Transformers, Prompt Engineering, Embeddings, RAG, Evaluation

RELEVANT COURSES

- Introduction to Programming (in C), Fundamentals of Scientific Computing, Computational Methods in Science.
- **Mathematics Courses:** Calculus I, Calculus II, Elementary Linear Algebra, Differential Equations.

ACHIEVEMENTS

- Solved 300+ DSA problems on platforms like LeetCode and Coding Ninjas.
- Secured A Grade in the NCC B-Certificate Examination and awarded SUO, the highest cadet rank at the institute.